

Example

$$\begin{cases} x_1 + 2x_2 - x_3 = 5 \\ x_1 + 2x_2 - 2x_3 = 4 \\ -2x_1 + x_2 + x_3 = 1 \end{cases}$$

a linear equation system. Find the solution of linear equation system by using LU-decomposition.

Solution

$$\underbrace{\begin{bmatrix} 1 & 2 & -1 \\ 1 & 2 & -2 \\ -2 & 1 & 1 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}_x = \underbrace{\begin{bmatrix} 5 \\ 4 \\ 1 \end{bmatrix}}_B$$

$$a_{11} = 1 \neq 0, \quad \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = \begin{vmatrix} 1 & 2 \\ 1 & 2 \end{vmatrix} = 2 - 1 = 1 \neq 0 \text{ and}$$

$$|A| = \begin{vmatrix} 1 & 2 & -1 \\ 1 & 2 & -2 \\ -2 & 1 & 1 \end{vmatrix} = 1 \cdot \begin{vmatrix} 2 & -2 \\ 1 & 1 \end{vmatrix} - 1 \cdot \begin{vmatrix} 1 & -2 \\ -2 & 1 \end{vmatrix} + (-1) \cdot \begin{vmatrix} 1 & 2 \\ -2 & 1 \end{vmatrix} = 2 \neq 0$$

So, A can be written as product of L and U such that $A = LU$

$$\begin{bmatrix} 1 & 2 & -1 \\ 1 & 2 & -2 \\ -2 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -1 \\ 1 & 2 & -2 \\ -2 & 1 & 1 \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} & l_{21}u_{13} + u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + u_{33} \end{bmatrix}$$

$$u_{11} = 1, \quad u_{12} = 1, \quad u_{13} = -1$$

$$l_{21}u_{11} = 1, \quad l_{21}u_{12} + u_{22} = 2, \quad l_{21}u_{13} + u_{23} = -2$$

$$l_{31}u_{11} = -2, \quad l_{31}u_{12} + l_{32}u_{22} = 1, \quad l_{31}u_{13} + l_{32}u_{23} + u_{33} = 1$$

$$\boxed{u_{11} = 1}$$

$$l_{21}u_{11} = 1 \Rightarrow \boxed{l_{21} = 1}$$

$$l_{31}u_{11} = -2 \Rightarrow \boxed{l_{31} = -2}$$

$$\boxed{u_{12} = 1}$$

$$l_{21}u_{12} + u_{22} = 2 \Rightarrow 1 \cdot 1 + u_{22} = 2$$

$$\Rightarrow \boxed{u_{22} = 1}$$

$$l_{31}u_{12} + l_{32}u_{22} = 1$$

$$-2 \cdot 1 + l_{32} \cdot 1 = 1$$

$$-2 + l_{32} = 1 \Rightarrow \boxed{l_{32} = 3}$$

$$\boxed{u_{13} = -1}$$

$$l_{21}u_{13} + u_{23} = -2$$

$$1 \cdot (-1) + u_{23} = -2 \Rightarrow -1 + u_{23} = -2 \Rightarrow \boxed{u_{23} = -1}$$

$$l_{31}u_{13} + l_{32}u_{23} + u_{33} = 1$$

$$-2 \cdot (-1) + 3 \cdot (-1) + u_{33} = 1$$

$$2 + (-3) + u_{33} = 1$$

$$\boxed{u_{33} = 2}$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -2 & 3 & 1 \end{bmatrix} \text{ and } U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix} = \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 2 \end{bmatrix}$$

2. way To find $A = L \cdot U$

$$\begin{bmatrix} 1 & 1 & -1 \\ 1 & 2 & -2 \\ -2 & 1 & 1 \end{bmatrix}$$

1. step: By applying elementary row operations we get a new A matrix as an upper triangular matrix

$$\begin{bmatrix} 1 & 1 & -1 \\ 1 & 2 & -2 \\ -2 & 1 & 1 \end{bmatrix} \xrightarrow[R_{31}(2)]{R_{21}(-1)} \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & -1 \\ 0 & 3 & -1 \end{bmatrix} \xrightarrow{R_{32}(-3)} \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 2 \end{bmatrix}$$

$\bar{E}_2 \cdot \bar{E}_1 \cdot A$ $\bar{E}_1 \cdot \bar{E}_2 \cdot \bar{E}_1 \cdot A = U$

$$\bar{E}_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_{21}(-1)} \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \bar{E}_1$$

$$\bar{E}_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \text{ and } \bar{E}_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix}$$

2. step: $\bar{E}_3 \cdot \bar{E}_2 \cdot \bar{E}_1 \cdot A = U$

$$\bar{E}_3^{-1} \cdot \bar{E}_2^{-1} \cdot \bar{E}_1^{-1} \cdot A = \bar{E}_3^{-1} \cdot U \Rightarrow A = \underbrace{\bar{E}_1^{-1} \cdot \bar{E}_2^{-1} \cdot \bar{E}_3^{-1}}_L \cdot U$$

$$\bar{E}_1 = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \bar{E}_1^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix}$$

$$\begin{aligned} l_{21} &= 1 \\ l_{31} &= -2 \\ l_{32} &= 3 \end{aligned}$$

$$\bar{E}_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \rightarrow \bar{E}_2^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$$

$$\bar{E}_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix} \rightarrow \bar{E}_3^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix}$$

$$\bar{E}_1^{-1} \cdot \bar{E}_2^{-1} \cdot \bar{E}_3^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -2 & 3 & 1 \end{bmatrix}}_L$$

So, A can be written as $A = L \cdot U$

$$\underbrace{\begin{bmatrix} 1 & 1 & -1 \\ 1 & 2 & -2 \\ -2 & 1 & 1 \end{bmatrix}}_A = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -2 & 3 & 1 \end{bmatrix}}_L \underbrace{\begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 2 \end{bmatrix}}_U$$

$$AX=B$$

$$\begin{bmatrix} 1 & 1 & -1 \\ 1 & 2 & -2 \\ -2 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \\ 1 \end{bmatrix}$$

$A=LU$

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \underbrace{\begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 2 \end{bmatrix}}_U \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}_X = \underbrace{\begin{bmatrix} 5 \\ 4 \\ 1 \end{bmatrix}}_B$$

L Y

1st step: Let $UX=Y$, we need to solve Y

$$L \cdot Y = B$$

$$\text{and } Y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

$$\Rightarrow L \cdot Y = B$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \\ 1 \end{bmatrix} \Rightarrow$$

$$y_1 = 5$$

$$y_1 + y_2 = 4 \Rightarrow y_2 = -1$$

$$-2y_1 + 3y_2 + y_3 = 1 \Rightarrow -2(5) + 3(-1) + y_3 = 1 \Rightarrow y_3 = 14$$

$$\text{we found } Y = \begin{bmatrix} 5 \\ -1 \\ 14 \end{bmatrix}$$

$$\text{Now } UX=Y$$

$$\begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5 \\ -1 \\ 14 \end{bmatrix} \Rightarrow$$

$$x_1 + x_2 - x_3 = 5$$

$$x_2 - x_3 = -1$$

$$2x_3 = 14 \Rightarrow x_3 = 7$$

$$x_2 - 7 = -1 \Rightarrow x_2 = 6$$

$$x_1 + 6 - 7 = 5 \Rightarrow x_1 = 6$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 6 \\ 6 \\ 7 \end{bmatrix}$$

Example

Let

$$2x + 2y + 2z = 12$$

$$4x + 7y + 7z = 24$$

$$6x + 18y + 22z = 12$$

be a linear equation system. By using LU decomposition find the

solution of linear equation system.

$$\underbrace{\begin{bmatrix} 2 & 2 & 2 \\ 4 & 7 & 7 \\ 6 & 18 & 22 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x \\ y \\ z \end{bmatrix}}_X = \underbrace{\begin{bmatrix} 12 \\ 24 \\ 12 \end{bmatrix}}_B$$

Firstly we check

$$a_{11} = 2 \neq 0, \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = \begin{vmatrix} 2 & 2 \\ 4 & 7 \end{vmatrix} = 6 \neq 0 \text{ and } \begin{vmatrix} 2 & 2 & 2 \\ 4 & 7 & 7 \\ 6 & 18 & 22 \end{vmatrix} = 2 \cdot \begin{vmatrix} 7 & 7 \\ 18 & 22 \end{vmatrix} - 2 \cdot \begin{vmatrix} 4 & 7 \\ 6 & 22 \end{vmatrix} + 2 \cdot \begin{vmatrix} 4 & 7 \\ 6 & 18 \end{vmatrix} \neq 0$$

So, matrix A can be written as product of L and U such that $A = LU$

1.4.4

$$\underbrace{\begin{bmatrix} 2 & 2 & 2 \\ 4 & 7 & 7 \\ 6 & 18 & 22 \end{bmatrix}}_A = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix}}_L \underbrace{\begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}}_U$$

$$\begin{bmatrix} 2 & 2 & 2 \\ 4 & 7 & 7 \\ 6 & 18 & 22 \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} & l_{21}u_{13} + u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + u_{33} \end{bmatrix}$$

$$\boxed{u_{11} = 2}$$

$$\boxed{u_{12} = 2}$$

$$\boxed{u_{13} = 2}$$

$$l_{21}u_{11} = 4$$

$$\boxed{l_{21} = 2}$$

$$l_{31}u_{11} = 6$$

$$\boxed{l_{31} = 3}$$

$$l_{21}u_{12} + u_{22} = 7$$

$$2 \cdot 2 + u_{22} = 7$$

$$\boxed{u_{22} = 3}$$

$$l_{31}u_{12} + l_{32}u_{22} = 18$$

$$3 \cdot 2 + l_{32} \cdot 3 = 18$$

$$3l_{32} = 12$$

$$\boxed{l_{32} = 4}$$

$$l_{21}u_{13} + u_{23} = 7$$

$$2 \cdot 2 + u_{23} = 7 \Rightarrow \boxed{u_{23} = 3}$$

$$l_{31}u_{13} + l_{32}u_{23} + u_{33} = 22$$

$$3 \cdot 2 + 4 \cdot 3 + u_{33} = 22 \Rightarrow \boxed{u_{33} = 4}$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 4 & 1 \end{bmatrix} \text{ and } U = \begin{bmatrix} 2 & 2 & 2 \\ 0 & 3 & 3 \\ 0 & 0 & 4 \end{bmatrix}$$

$$\underbrace{\begin{bmatrix} 2 & 2 & 2 \\ 4 & 7 & 7 \\ 6 & 18 & 22 \end{bmatrix}}_A = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 4 & 1 \end{bmatrix}}_L \underbrace{\begin{bmatrix} 2 & 2 & 2 \\ 0 & 3 & 3 \\ 0 & 0 & 4 \end{bmatrix}}_U$$

2. way

3. step: Applying elementary row operations we get upper triangular matrix U

$$\begin{bmatrix} 2 & 2 & 2 \\ 4 & 7 & 7 \\ 6 & 18 & 22 \end{bmatrix} \xrightarrow[\substack{R_{21}(-2) \\ R_{31}(-3)}]{} \begin{bmatrix} 2 & 2 & 2 \\ 0 & 3 & 3 \\ 0 & 12 & 16 \end{bmatrix} \xrightarrow{R_{32}(-4)} \begin{bmatrix} 2 & 2 & 2 \\ 0 & 3 & 3 \\ 0 & 0 & 4 \end{bmatrix} = U$$

E_2, E_1, A E_3, E_2, E_1, A

$$\bar{E}_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_{21}(-2)} \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \bar{E}_1$$

$$\bar{E}_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_{31}(-3)} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix} = \bar{E}_2$$

$$\bar{E}_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_{32}(-4)} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -4 & 1 \end{bmatrix} = \bar{E}_3$$

2. step $\bar{E}_3, \bar{E}_2, \bar{E}_1, A = U \Rightarrow A = \underbrace{\bar{E}_1^{-1} \cdot \bar{E}_2^{-1} \cdot \bar{E}_3^{-1}}_L \cdot U$

$$\bar{E}_1^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \bar{E}_2^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix}, \bar{E}_3^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 4 & 1 \end{bmatrix}$$

$$L = \bar{E}_1^{-1} \bar{E}_2^{-1} \bar{E}_3^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 4 & 1 \end{bmatrix}$$

So, $A = L \cdot U$

$$\begin{bmatrix} 2 & 2 & 2 \\ 4 & 7 & 7 \\ 6 & 18 & 22 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 4 & 1 \end{bmatrix} \begin{bmatrix} 2 & 2 & 2 \\ 0 & 3 & 3 \\ 0 & 0 & 4 \end{bmatrix}$$

A L U

Then, we need to solve system.

$$AX = B$$

$$(LU)X = B$$

$$L(UX) = B$$

$$\Rightarrow LY = B \text{ such that } UX = Y, \text{ where } Y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

$$1. y = B$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 4 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 12 \\ 24 \\ 12 \end{bmatrix} \Rightarrow \begin{array}{l} y_1 = 12 \\ 2y_1 + y_2 = 24 \Rightarrow y_2 = 0 \\ 3y_1 + 4y_2 + y_3 = 12 \Rightarrow 36 + 0 + y_3 = 12 \Rightarrow y_3 = -24 \end{array}$$

$$y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 12 \\ 0 \\ -24 \end{bmatrix}$$

Then, we next solve $UX = y$

$$\begin{bmatrix} 2 & 2 & 2 \\ 0 & 3 & 3 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 12 \\ 0 \\ -24 \end{bmatrix}$$

$$\begin{array}{l} -4z = 24 \Rightarrow z = -6 \\ 3y + 3z = 0 \Rightarrow 3y + 3(-6) = 0 \Rightarrow y = 6 \\ 2x + 2y + 2z = 12 \Rightarrow 2x + 12 + 2(-6) = 12 \\ \Rightarrow 2x = 12 \\ \Rightarrow x = 6 \end{array}$$

We have

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 6 \\ 6 \\ -6 \end{bmatrix}$$

Example
 Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & -1 & 4 \\ -1 & 2 & 1 \end{bmatrix}$. Write A as a product lower triangular matrix L and upper triangular matrix U .

Solution
 $A = L \cdot U$

$a_{11} = 1 \neq 0$, $\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{12} \end{vmatrix} = \begin{vmatrix} 1 & 2 \\ 2 & -1 \end{vmatrix} = -5 \neq 0$ and also $|A| \neq 0$, so A can be written as $L \cdot U$

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & -1 & 4 \\ -1 & 2 & 1 \end{bmatrix} \xrightarrow{R_{21}(-2)} \begin{bmatrix} 1 & 2 & 3 \\ 0 & -5 & -2 \\ -1 & 2 & 1 \end{bmatrix} \xrightarrow{R_{31}(1)} \begin{bmatrix} 1 & 2 & 3 \\ 0 & -5 & -2 \\ 0 & 4 & 4 \end{bmatrix} \xrightarrow{R_{32}(\frac{4}{5})} \begin{bmatrix} 1 & 2 & 3 \\ 0 & -5 & -2 \\ 0 & 0 & 12/5 \end{bmatrix} = U$$

$E_1 \cdot A$ $E_2 \cdot E_1 \cdot A$ $E_3 \cdot E_2 \cdot E_1 \cdot A = U$

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_{21}(-2)} \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = E_1$$

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_{31}(1)} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} = E_2$$

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_{32}(4/5)} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 4/5 & 1 \end{bmatrix} = E_3$$

2. step $E_3 \cdot E_2 \cdot E_1 \cdot A = U$

$$A = E_3^{-1} \cdot E_2^{-1} \cdot E_1^{-1} \cdot U$$

$$E_1^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad E_2^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \quad \text{and} \quad E_3^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -4/5 & 1 \end{bmatrix}$$

$$E_1^{-1} \cdot E_2^{-1} \cdot E_3^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -4/5 & 1 \end{bmatrix} = L$$

$$\underbrace{\begin{bmatrix} 1 & 2 & 3 \\ 2 & -1 & 4 \\ -1 & 2 & 1 \end{bmatrix}}_A = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -4/5 & 1 \end{bmatrix}}_L \underbrace{\begin{bmatrix} 1 & 2 & 3 \\ 0 & -5 & -2 \\ 0 & 0 & 12/5 \end{bmatrix}}_U$$